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An advanced kinetic model of electron-beam-excited KrF lasers including the vibrational relaxation in KrF*(B) and collisional mixing of KrF*(B,C)

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Computer models developed so far on electron-beam-excited KrF(B–X, 248 nm) lasers that include the vibrational relaxation process in the upper lasing B level at the finite rate could not predict the high intrinsic laser efficiency which was experimentally reported. This is attributed to the reduction of the laser extraction efficiency. We have developed a four-level KrF laser model that includes the vibrational relaxation process and also the collisional mixing of the KrF*(B) and the KrF*(C) levels. The collisional quenching rates for KrF*(B,C) that we used and the vibrational relaxation rate were carefully estimated by using the effective spontaneous lifetimes for KrF*(B,C). As a result, the model prediction was in quite good agreement with many experimental results for a saturation behavior of KrF*(B–X) fluorescence, for small-signal gains, for small-signal absorptions, and for intrinsic efficiencies. Estimated rate constants in this model for the vibrational relaxation and the KrF*(B,C) mixing are 4×10^{-11} and 5×10^{-10} cm^3/s, respectively, for a two-body collision rate with argon gas.

I. INTRODUCTION

The electron-beam-excited KrF laser is the highest power and the highest-efficiency laser in the UV region of the spectrum. It is also one of the prime candidates for an advanced Inertial Confinement Fusion (ICF) energy driver.[1,2] Because the KrF laser is a non-storage-type laser, the additional pulse compression scheme is essential for a practical use in the ICF.[3,4] Since Tisone, Patterson, and Rice[5–7] demonstrated an extremely high intrinsic efficiency of 12% with a short e-beam pulse (60-nm FWHM) pumping at an excitation rate of 7 MW/cm^3 and at a total laser gas pressure of 2500 Torr, the direct production of laser pulses less than 100 ns using a high excitation rate pumping has been actively investigated.[8–12] However, large amplifier modules must be practically designed from such scaling criteria as the intrinsic laser efficiency, complexity, and cost of optical system involved in the angular multiplexing, pulsed power technology for the e-beam generator, loading on the optical components, pressure in the amplifier cell, the number of amplifier stages, and the specific laser energy. Current conceptual amplifier designs are done with a long-pulse (200–500-ns) and low-excitation-rate (0.1–0.2 MW/cm^3) pumping scheme along with the angular multiplexing.[2,13] On the other hand, as a result of theoretical KrF laser research, laser performance has been widely understood in terms of various excitation rates and laser gas mixtures.[1,2,12,14–19] This progress has resulted in some conceptual designs of several tens kJ KrF laser amplifiers.[1,2]

Recently, however, Jacob *et al.*[20] reported that because of the finite rate of vibrational relaxation in the upper KrF laser B state, some of the KrF* population in the higher vibrational levels cannot be efficiently extracted by stimulated emission, which results in the decrease in the KrF laser efficiency. This effect should be included in the kinetic model because all models so far treated the KrF laser as a one-level laser using the assumption of an infinite vibrational relaxation rate. There is some experimental evidence that the vibrational relaxation process is not as fast as the lasing process. Therefore, one cannot treat the model as the one-state, upper-level model. Jacob *et al.*[21] measured the sidelight fluorescence (B–X) signals observed in the presence and absence of the cavity flux, and found that this fluorescence was not saturated as predicted from the one-state, upper-level model. By comparing the steady-state analysis of fluorescence saturation behavior with the new two-state, upper-level-model, they estimated the finite vibrational relaxation rate constant to be 5×10^{-11} cm^3/s for a two-body collision rate with Ar gas.[21] The two-state, upper-level model includes two vibrational states lying at the KrF*(B) level. Similar experiments were reported by Mandel and co-workers.[22] They estimated a vibrational relaxation rate constant of 4×10^{-11} cm^3/s. Grattan and co-workers[23] reported the dynamic gain recovery experiment on the 248-nm KrF laser amplifier using a short KrF laser probe pulse of about 4.7 ns with the saturation intensity. Comparing the experimental results of Gratton and co-workers[23] with the simple rate-equation calculation, they determined the vibrational relaxation rate constant of $\geqslant 2.5\times10^{-11}$ cm^3/s for argon gas. Discharge-excited KrF lasers using He buffer gas were also used in some experiments. Pummer and co-workers[24] observed the sidelight fluorescence and clarified the nonequilibrium relaxation in the KrF* vibrational level population. They estimated a vibrational relaxation time constant of 4 ± 2 ns from the decay time of fluorescence at a total pressure of about 2 atm. Szatmari and Schafer[25] measured the picosecond, dynamic-gain behavior using a probe laser where a high-power and short-pulse (<10 ps) laser was injected into the KrF amplifier. They concluded that as a result of the vibrational relaxation in both the KrF*(B) state and the

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$KrF^*(C)$ state and of the repumping by the discharge, the long gain recovery time is 4 ns at a He-buffered mixture pressure of about 2.5 atm. A detailed kinetic calculation of the vibrational relaxation in $KrF^*(B)$ state in a mixture of Kr and F_2 was reported by Morgan and co-workers.[26] They constructed a collisional-radiative model consisting of rate equations for twenty discrete vibrational state $KrF(v=0)$ through $KrF(v=19)$. Although their calculation was only for the laser mixture of Kr/F_2 without Ar buffer, they have shown that rapid vibrational relaxation by symmetric charge transfer is still slow enough for excitation to be lost by spontaneous emission and collisional quenching. This leads to a significant reduction in the predicted laser extraction efficiency.

Every experimental result on the vibrational relaxation shows the finite rate constants of 2×10^{-11}–5×10^{-11} cm^3/s for a two-body collision rate constant with Ar buffer gas. However, by introducing the two-state, upper-level model developed by Jacob *et al.*[21] into the laser kinetic model, the prediction by using a vibrational relaxation rate constant of 5×10^{-11} cm^3/s showed a significant reduction of laser intrinsic efficiency of 25%–50% from that calculated by the one-state, upper-level model.[15,17,22] The reduction depends on both the total pressure of the laser mixture and excitation rate. Therefore, the higher intrinsic efficiency of around 10% which was experimentally obtained can be predicted only by either the one-state, upper-level model or the two-state, upper-level model with a very fast vibrational relaxation rate constant of in excess of 1×10^{-9} cm^3/s. Therefore, we must improve the kinetic model so as to agree with the experiments in terms of the fluorescence efficiencies, small-signal gains, small-signal absorptions, and intrinsic efficiencies which have been explained by a simple one-state, upper-level model, and the vibrational relaxation behavior.

We solved this issue by introducing not only the slow vibrational relaxation effect in the $KrF^*(B)$ state but also the dynamic mixing between the $KrF^*(B)$ state and the $KrF^*(C)$ state. Relative to simple kinetic models as previous models,[14,15] novel points are summarized.

(1) The simple point source model is extended to the one-dimensional model along the optical axis. Therefore, the propagation of photon in the cavity is treated in a real manner.

(2) The upper laser level is treated as a set of four states including the vibrational relaxation processes in the $KrF^*(B)$ and the $KrF^*(C)$ state, which is called a four-state, upper-level model.

(3) Assuming that the fast mxing is done between the $KrF^*(B)$ and the $KrF^*(C)$ state, the effective radiative lifetime of $KrF^*(B,C)$ can be calculated and the previous collisional quenching rate constants are reexamined.

(4) ArF^* as well as KrF^* is also treated as the mixing of the $ArF^*(B)$ state with the $ArF^*(C)$ state. From the effective lifetime, the rate constants of the quenching reactions and the substitutive reaction of $ArF^* + Kr \rightarrow KrF^* + Ar$ are reexamined.

(5) Three-body ion–ion recombination rates are treated as a function of instantaneous gas temperature, and the plasma-shielding effects at high ion densities of the order of 10^{13}–10^{15} cm^3 are included.

(6) The temporal secondary electron temperature is calculated assuming that the secondary electron energy distribution is Maxwellian, and reaction rate constants are adjusted temporally.

The performance of the new model was widely checked by comparing the predictions with small-signal gains and small-signal absorptions, laser extraction characteristics, intrinsic efficiencies, and the saturation behavior of sidelight fluorescence reported previously by various experimenters,[5–7] and good agreement was obtained. Furthermore, when this model was compared with our experimental results for the temporal behavior of sidelight fluorescence using 50-ns *e*-beam excitation, good agreement was also attained.

II. CODE DEVELOPMENT

A. One-dimensional treatment

The baseline of the numerical model in this paper is the same one as that reported in Refs. 14 and 15 and basically similar to the *e*-beam-excited XeCl laser model.[27] All of the reaction processes and their rate constants involved in the model for a mixture of $Ar/Kr/F_2$ are shown in Table I, which is based on the careful collection of rate constants by Johnson and Hunter.[16] In order to calculate the time history of sidelight fluorescence observed at the center of the laser cell, we improved the model of laser extraction process by using a one-dimensional treatment. We ignored the transverse variations. The photon transport equations are written as

$$\frac{\partial\phi_\pm}{\partial t} \pm c\frac{\partial\phi_\pm}{\partial x} = c\left(\frac{g_0-\alpha_0}{1+(\phi_+ + \phi_-)/\phi_s} - \alpha\right)\phi_\pm + S\Delta\Omega/4\pi, \tag{1}$$

where x denotes axial position along the laser axis and t is time. The velocity of light is c and ϕ is the intracavity laser flux. Superscript $\pm$ indicates the direction of photon propagation. Parameters g_0, α_0, and α are small-signal gain, small-signal saturable absorption, and nonsaturable absorption coefficient, respectively. Parameter S denotes the spotaneous emission term, and $\Delta\Omega$ is the solid angle calculated from the laser cell geometry. This model equation takes the form of a hyperbolic partial differential equation. This can be recast as ordinary differential equations, if the differentials are taken along the characteristics. The equation has the general form:

$$\frac{d\phi_\pm}{dt} = c\left(\frac{g_0-\alpha_0}{1+(\phi_+ + \phi_-)/\phi_s} - \alpha\right)\phi_\pm + S\Delta\Omega/4\pi \quad \text{along } dx/dt = \pm c. \tag{2}$$

The rate equations for number densities of various species and electrons are of the form:

$$\frac{dn(x,t)}{dt} = \text{CHEM} - \sum \text{RAD}, \text{ along } dx/dt = 0 \tag{3}$$

where CHEM denotes the chemical reactions and RAD denotes the radiative processes. The model includes the number density variation along the optical axis only for the saturable species $KrF^*(B)$, $KrF^*(C)$, Kr_2F^*, and $ArKrF^*$. For

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TABLE I. KrF^* kinetic rate constants used in the model. Units are in cm^3/s unless otherwise noted.

Reaction	Rate constant	Ref.
Neutral reaction		
$Ar^* + F_2 \rightarrow ArF^* + F$	$7.5(-10)$	52,59,92
$Ar^* + Kr + Ar \rightarrow ArKr^* + Ar$	$1.0(-32)\ cm^6/s$	52
$Ar^* + 2Ar \rightarrow Ar_2^* + Ar$	$1.14(-32)\ cm^6/s$	52
$Ar^* + Kr \rightarrow Ar + Kr^*$	$6.2(-12)$	52,60
$Ar^* + F_2 \rightarrow Ar + 2F$	$3.1(-10)$	92
$Ar^{**} + F_2 \rightarrow ArF^* + F$	$4.7(-10)$	52,60
$Ar^{**} + Ar \rightarrow Ar^* + Ar$	$1.0(-10)$	estimated
$Ar^{**} + F_2 \rightarrow Ar + 2F$	$3.1(-10)$	analogous to Ar^*
$Ar_2^* + F_2 \rightarrow Ar_2F^* + F$	$2.5(-10)$	52
$Ar_2^* + F \rightarrow ArF^* + Ar$	$3.0(-10)$	52
$Ar_2^* + Kr \rightarrow Kr^* + 2Ar$	$4.0(-10)$	43
$Ar_2^* + F_2 \rightarrow ArF^* + Ar + F$	$3.0(-10)$	61
$Kr^* + F_2 \rightarrow KrF^* + F$	$8.1(-10)$	52,59,93
$Kr^* + 2Ar \rightarrow ArKr^* + Ar$	$1.0(-32)\ cm^6/s$	52
$Kr^* + Kr + Ar \rightarrow Kr_2^* + Ar$	$2.7(-32)\ cm^6/s$	52
$Kr^{**} + F_2 \rightarrow KrF^* + F$	$8.1(-10)$	52
$Kr^{**} + Ar \rightarrow Kr^* + Ar$	$1.0(-10)$	estimated
$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	$3.0(-10)$	52
$Kr_2^* + F_2 \rightarrow KrF^* + Kr + F$	$3.0(-10)$	61
$Kr_2^* + F \rightarrow KrF^* + Kr$	$3.0(-10)$	52
$ArKr^* + F_2 \rightarrow KrF^* + Ar + F$	$6.0(-10)$	52
$ArKr^* + Kr \rightarrow Kr_2^* + Ar$	$1.0(-10)$	52
$ArKr^* + F_2 \rightarrow ArKrF^* + F$	$3.0(-10)$	analogous to Ar_2^*
$F + F + M \rightarrow F_2 + M$	$1.0(-33) cm^6/s$	16
ArF^* relaxation		
$ArF^* + Kr \rightarrow KrF^* + Ar$	$1.6(-09)$	62[a]
$ArF^* + 2Ar \rightarrow Ar_2F^* + Ar$	$4.0(-31) cm^6/s$	62,63,90[a,b]
$ArF^* + F_2 \rightarrow Ar + F + F_2$	$1.9(-9)$	62,63[a]
$ArF^* + Ar \rightarrow 2Ar + F$	$9.0(-12)$	62,63[a]
Ar_2F^* relaxation		
$Ar_2F^* + F_2 \rightarrow 2Ar + F + F_2$	$2.0(-10)$	61
$Ar_2F^* + Kr \rightarrow ArKrF^* + Ar$	$1.0(-10)$	52,59
$Ar_2F^* + Kr \rightarrow KrF^* + 2Ar$	$1.0(-10)$	estimated
$Ar_2F^* + Ar \rightarrow 3Ar + F$	$1.0(-10)$	estimated
KrF^* relaxation		
$KrF^* + Kr \rightarrow 2Kr + F$	$2.0(-12)$	estimated[a]
$KrF^* + Ar \rightarrow Kr + Ar + F$	$1.8(-12)$	estimated[a]
$KrF^* + 2Ar \rightarrow ArKrF^* + Ar$	$7.0(-32)\ cm^6/s$	62,63[a,b]
$KrF^* + Kr + Ar \rightarrow Kr_2F^* + Ar$	$6.5(-31) cm^6/s$	62,63[a,b]
$KrF^* + F_2 \rightarrow Kr + F + F_2$	$7.8(-10)$	62,63[a]
$KrF^* + 2Kr \rightarrow Kr_2F^* + Kr$	$6.7(-31) cm^6/s$	62,63[a,b]
$KrF^* + F \rightarrow Kr + 2F$	$7.8(-10)$	analogous to F_2[a]
Kr_2F^* relaxation		
$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	$1.5(-10)$	89,94
$ArKrF^*$ relaxation		
$ArKrF^* + F_2 \rightarrow Ar + Kr + F + F_2$	$1.0(-09)$	52
$ArKrF^* + Kr \rightarrow Kr_2F^* + Ar$	$2.0(-11)$	43
$ArKrF^* + Ar \rightarrow Ar_2F^* + Kr$	$2.0(-11)$	43
Electron excitation, ionization[72] and attachment		
$Ar + e \leftrightarrow Ar^* + e$	$5.0(-09) Te^{0.74} \exp(-11.56/Te)$	73,74,75
$Ar + e \leftrightarrow Ar^{**} + e$	$1.40(-80) Te^{0.71} \exp(-13.15/Te)$	73,74,75
$Ar^* + e \leftrightarrow Ar^{**} + e$	$8.90(-07) Te^{0.51} \exp(-1.590/Te)$	76
$Ar + e \rightarrow Ar^+ + 2e$	$2.30(-08) Te^{0.68} \exp(-15.76/Te)$	77,78
$Ar^* + e \rightarrow Ar^+ + 2e$	$6.80(-09) Te^{0.67} \exp(-4.20/Te)$	79
$Ar^{**} + e \rightarrow Ar^+ + 2e$	$1.80(-07) Te^{0.61} \exp(-2.61/Te)$	79
$Ar_2^* + e \rightarrow Ar_2^+ + 2e$	$9.00(-08) Te^{0.70} \exp(-3.66/Te)$	72
$Kr + e \leftrightarrow Kr^* + e$	$9.00(-09) Te^{0.72} \exp(-9.96/Te)$	73,74,75
$Kr + e \leftrightarrow Kr^{**} + e$	$2.30(-08) Te^{0.72} \exp(-11.47/Te)$	73,74,75
$Kr^* + e \leftrightarrow Kr^{**} + e$	$9.00(-07) Te^{0.50} \exp(-1.51/Te)$	76
$Kr + e \rightarrow Kr^+ + 2e$	$2.70(-08) Te^{0.70} \exp(-14.00/Te)$	77,78
$Kr^* + e \rightarrow Kr^+ + 2e$	$7.00(-08) Te^{0.68} \exp(-4.04/Te)$	79
$Kr^{**} + e \rightarrow Kr^+ + 2e$	$2.00(-07) Te^{0.62} \exp(-2.53/Te)$	79
$Kr_2^* + e \rightarrow Kr_2^+ + 2e$	$9.00(-08) Te^{0.70} \exp(-3.53/Te)$	72
$ArKr^* + e \rightarrow ArKr^+ + 2e$	$9.00(-08) Te^{0.70} \exp(-3.60/Te)$	72

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TABLE I. (Continued)

Reaction	Rate constant	Ref.
$F_2 + e - \text{beam} \rightarrow F_2^+ + e - \text{beam} + e$:	32 eV/ion electron	
$Ar + e - \text{beam} \rightarrow Ar^+ + e - \text{beam} + e$:	26.2 eV/ion electron	
$Kr + e - \text{beam} \rightarrow Kr^+ + e - \text{beam} + e$:	24.3 eV/ion electron	
$Ar + e - \text{beam} \rightarrow Ar^* + e - \text{beam}$:	0.28 × ionization rate	
$Kr + e - \text{beam} \rightarrow Kr^* + e - \text{beam}$:	0.28 × ionization rate	
$F_2 + e \rightarrow F^- + F$	4.5(−09) ($Te = 1$ eV)	17
Penning ionization		
$Ar^{**} + Ar^{**} \rightarrow Ar^+ + Ar + e$	5.0(−10)	estimated
$Ar^* + Ar^* \rightarrow Ar^+ + Ar + e$	5.0(−10)	86
$Ar_2^* + Ar_2^* \rightarrow Ar_2^+ + 2Ar + e$	5.0(−10)	87
$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e$	5.0(−10)	87
$Kr^* + Kr^* \rightarrow Kr^+ + Kr + e$	5.0(−10)	86
Charge transfer reaction		
$Ar_2^* + Kr \rightarrow Kr^+ + 2Ar$	7.5(−10)	64
$Ar^+ + Kr \rightarrow Kr^+ + Ar$	3.0(−11)	64
$Ar^+ + 2Ar \rightarrow Ar_2^+ + Ar$	2.5(−31) cm^6/s	65
$Ar^+ + Kr + Ar \rightarrow ArKr^+ + Ar$	1.0(−31) cm^6/s	66
$Kr^+ + 2Ar \rightarrow ArKr^+ + Ar$	1.0(−31) cm^6/s	16
$ArKr^+ + Kr \rightarrow Kr_2^+ + Ar$	3.2(−10)	16
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.5(−31) cm^6/s	65
$Kr^+ + Ar + Kr \rightarrow Kr_2^+ + Ar$	2.5(−31) cm^6/s	43,67
Three-body ion–ion recombination reaction		
$Ar^+ + F^- + (Ar, Kr) \rightarrow ArF^* + (Ar, Kr)$		
$Ar_2^+ + F^- + (Ar, Kr) \rightarrow ArF^* + Ar + (Ar, Kr)$		
$ArKr^+ + F^- + (Ar, Kr) \rightarrow KrF^* + (Ar, Kr)$	(cm^6/s)	28–35
$ArKr^+F^- + (Ar, Kr) \rightarrow ArKrF^* + (Ar, Kr)$		see text
$Kr^+ + F^- + (Ar, Kr) \rightarrow KrF^* + (Ar, Kr)$		
$Kr_2^+ + F^- + (Ar, Kr) \rightarrow KrF^* + Kr + (Ar, Kr)$		
$F_2^+ + F^- \rightarrow 3F$	4.0(−8)	91
Desociative recombination reaction		
$Ar_2^+ + e \rightarrow Ar^* + Ar$	0.6(−6)$[Te(K)/300]^{-0.66}$	71
$Ar_2^+ + e \rightarrow Ar^{**} + Ar$	1.1(−07)	68,69,70
$Kr_2^+ + e \rightarrow Kr^* + Kr$	1.2(−6)$[Te(K)/300)^{-0.55}$	71
$Kr_2^+ + e \rightarrow Kr^{**} + Kr$	1.9(−07)	68,69,70
$ArKr^+ + e \rightarrow Kr^{**} + Ar$	1.2(−6)$[Te(K)/300]^{-0.55}$	71
Superelastic reaction		
$ArF^* + e \rightarrow Ar + F + e$	1.6(−07)	45[a]
$Ar_2F^* + e \rightarrow 2Ar + F + e$	1.0(−07)	17
$Kr_2F^* + e \rightarrow 2Kr + F + e$	1.0(−07)	estimated
$KrF^* + e \rightarrow Kr + F + e$	2.0(−07)	45[a]
$ArKr^* + e \rightarrow Ar + Kr + e$	1.0(−07)	estimated
$ArKrF^* + e \rightarrow Ar + Kr + F + e$	1.0(−07)	estimated
$Ar_2^* + e \rightarrow 2Ar + e$	1.0(−07)	estimated
$Kr_2^* + e \rightarrow 2Kr + e$	1.0(−07)	estimated
Radiative lifetime		
$KrF^*(B) \rightarrow Kr + F + h\nu$	1.43(08) s^{-1}	47,48
$KrF^*(C) \rightarrow Kr + F + h\nu$	1.33(07) s^{-1}	47,48
$ArF^*(B) \rightarrow Ar + F + h\nu$	2.50(08) s^{-1}	47,48
$ArF^*(C) \rightarrow Ar + F + h\nu$	2.27(07) s^{-1}	47,48
$Kr_2^* \rightarrow 2Kr + h\nu$	8.0(07) s^{-1}	88
$Ar_2^* \rightarrow 2Ar + h\nu$	6.0(07) s^{-1}	88
$Kr_2F^* \rightarrow 2Kr + F + h\nu$	5.6(06) s^{-1}	90,93
$Ar_2F^* \rightarrow 2Ar + F + h\nu$	5.4(06) s^{-1}	61,93
$ArKr^* \rightarrow Ar + Kr + h\nu$	8.0(07) s^{-1}	estimated
$ArKrF^* \rightarrow Ar + Kr + F + h\nu$	5.0(06) s^{-1}	estimated
248-nm absorption cross section	(cm^2)	
$F_2 + h\nu \rightarrow 2F$	1.2(−20)	81
$F^- + h\nu \rightarrow F + e$	5.6(−18)	82
$Kr_2^+ + h\nu \rightarrow Kr^+ + Kr$	1.6(−18)	62
$Ar_2^+ + h\nu \rightarrow Ar^+ + Ar$	1.3(−17)	83
$Ar^{**} + h\nu \rightarrow Ar^+ + e$	6.0(−18)	84
$Kr^{**} + h\nu \rightarrow Kr^+ + e$	6.0(−18)	84
$Ar^* + h\nu \rightarrow Ar^+ + e$	4.5(−18)	16

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TABLE I. (Continued)

Reaction	Cross section (cm^2)	Ref.
$Kr^* + h\nu \rightarrow Kr^+ + e$	1.0(−19)	16
$Kr_2F^* + h\nu \rightarrow 2Kr + F$	6.0(−18)	16
$Ar_2F^* + h\nu \rightarrow 2Ar + F$	1.0(−18)	16
$KrF^* + h\nu \rightarrow Kr + F + 2h\nu$	2.6(−16)	46
$ArKr^* + h\nu \rightarrow Ar^+ + Kr + h\nu$	1.5(−17)	85

[a] These quenching rate constants were experimentally evaluated from the product of quenching rate constant k and radiative lifetime of $RgX^*(B)$ state $\tau(B)$. Considering the mixing process of $RgX^*(B)/(C)$ at a finite rate, these quenching rate constants were modified using the effective radiative lifetime, $\tau(B/C)$. Reduction factors for ArF^* quenching rate constants and KrF^* quenching rate constants are 0.418 and 0.433, respectively.

[b] Expressed for 300 K gas temperature. Temperature dependence is available from the calculation method in Refs. 49–51.

the other species, $n(x,t) = n(t)$. In our model, the cavity distance is divided into $\Delta x = c\Delta t$. Therefore, for example, $\phi_+(x,t+\Delta t)$ is calculated by $\phi_+(x-\Delta x,t)$ and $\phi_-(x+\Delta x,t)$ using Eq. (2). A set of Eqs. (2) and (3), the energy balance equation for calculating the instantaneous gas temperature, and the electron energy balance equation are solved by a fourth-order Runge–Kutta method.

The effect of the one-dimensional treatment on the laser efficiency is not as appreciable as the zero-dimensional treatment for the typical operational conditions. However, when laser windows have large optical losses and optical absorption by F_2 gas in nonexcitation region is large, there appears to be a difference between two models. Furthermore, in Eq. (2), we introduced the spontaneous term, where the same spectral width of laser as the spontaneous emission is considered. Therefore in the $\phi_\pm$, the contribution of $ASE_\pm$ (amplified spontaneous emission) is included. The contribution of ASE radiated from the nonextracted volume cannot be involved in the calculation. In fact, under conditions where the laser gain is loaded by the laser flux, this spontaneous emission term can be ignored. Therefore, to calculate laser oscillator performance, a spontaneous emission term acts only as a seed of the stimulated emission. Parasitics from the side walls of the laser chamber could dominate the process. Therefore, we treated the side walls to reduce the reflections.

B. Three-body ion–ion recombination

In the previous models, the rate constants for ion–ion recombination reactions reported by Wadehra and Bardsley,[28] Flannery and Yang,[29–32] and Hoffman and Moreno[33] which are written as effective two-body collisional rate constants derived from the modified Natanson calculation, have been widely used. A Monte Carlo calculation to get the three body rate constant was also reported by Bardsley and Wadehra[34] where fairly good agreement was obtained with Wadehra's calculation by Wadehra and Bardsley.[28] At ion densities from 10^{13} to 10^{15} cm^{-3}, the effects of other ions on the recombination rate should be taken into account by using a Debye screened potential rather than a Coulomb potential. This is a plasma shielding effect first proposed by Morgan and co-workers.[35] They calculated the rate constant for $Kr^+ + F^-$ in Ar buffer gas as a function of Ar gas pressure for parameters of ion density using a Monte Carlo method. According to their results, the plasma shielding effect clearly decreased the recombination rate constant. Therefore, in our new model, in terms of the total pressure and the gas-temperature-dependent recombination rate, the modified Natanson calculation[33] is directly introduced and their profile is used. However, the absolute values of recombination rates are calculated using the reduction factor which is calculated by comparing Morgan's result[35] with Hoffman's result.[33] Because Morgan's calculation is only for the reaction $Kr^+ + F^- + Ar$, we simply assumed that the reduction factor is constant for every recombination reaction. For example, at an ion density of 10^{14} cm^{-3} in 2-atm Ar gas, the reduction factor is 0.397. However, because the recombination rate is faster than other reaction rates, this reduction does not cause the excimer formation efficiency to appreciably decrease through the ion channels.

C. Four-state, upper-level model

Dynamic mixing of the B state with the C state has already been introduced into the computer models of XeF laser[36] and XeCl lasers.[37,38] Because the transition spectrum of $KrF^*(C–A)$ and Ar_2F^* overlaps each other, the relative intensities of $KrF^*(B–X)$ to $KrF^*(C–A)$ fluorescence in Ar buffer cannot be measured. This is one of the reasons why the $KrF^*(B)$ state mixing with the $KrF^*(C)$ state has not, so far, been a kinetic issue to be solved. But the $KrF^*(C)$ state can be neglected only in the case where the Boltzmann factor of the C state is much less than that of the B state, and the mixing process is extremely fast. In fact, the energy gap $\Delta E = E_B - E_C$ between the B state and the C state calculated from the results reported by Koltz and Setser[39] is about 120 cm^{-1}. Recently, Julienne and Krauss[40] reexamined Koltz's results by taking account of $KrF^*(B–A)$ transition and then obtained $\Delta E = E_B - E_C = 80$ cm^{-1}. Therefore, under the steady-state Boltzmann distribution characterized by the gas temperature, the number density of KrF^* in the B state and the C state can be nearly equal. Even if the $KrF^*(B)$ state and the (C) state are treated as a single state on the assumption of their rapid mixing, the radiative lifetime of the mixed state should be an appropriate statistically weighted average of the lifetimes for the two states. We treated this mixing process using the finite mixing rate. As a result of the combination of this dynamic mixing model of the $KrF^*(B)$ state and the $KrF^*(C)$ state, and the vibrational relaxation model in these states, the upper level set is described by four equations:

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$$\frac{dN_{Bv_i}}{dt} = -\frac{N_{Bv_i}-\theta_{v_i}N_B}{\tau_v} - \frac{N_{Bv_i}-\theta_B(N_{Bv_i}+N_{Cv_i})}{\tau_{BC}} - \frac{N_{Bv_i}}{\tau_B} - \frac{N_{Bv_i}}{\tau_s} - \frac{N_{Bv_i}}{\tau_Q} (i=0-4), \quad (4)$$

$$\frac{dN_{Cv_i}}{dt} = -\frac{N_{Cv_i}-\theta_{v_i}N_C}{\tau_v} - \frac{N_{Cv_i}-\theta_C(N_{Cv_i}+N_{Cv_i})}{\tau_{BC}} - \frac{N_{Cv_i}}{\tau_c} - \frac{N_{Cv_i}}{\tau_Q} (i=0-4), \quad (5)$$

$$\frac{dN_B}{dt} = 0.5R - \frac{N_B-\theta_B(N_B+N_c)}{\tau_{BC}} - \frac{N_B}{\tau_B} - \frac{\Sigma N_{Bv_i}}{\tau_s} - \frac{N_B}{\tau_Q}, \quad (6)$$

and

$$\frac{dN_C}{dt} = 0.5R - \frac{N_C-\theta_C(N_B+N_C)}{\tau_{BC}} - \frac{N_C}{\tau_C} - \frac{N_C}{\tau_Q}, \quad (7)$$

where N_{Bv_i} and N_{Cv_i} denote the KrF* number density in ith vibrational level of the KrF*(B) state and the KrF*(C) state, respectively. Also, $N_B = \Sigma N_{Bv_i}$; $N_C = \Sigma N_{Cv_i}$; θ_{v_i} is the Boltzmann factor for the ith vibrational level; θ_B and θ_C are the Boltzmann factors for the KrF*(B) and the KrF*(C) states, respectively; and quantities τ_B, τ_C, τ_s, and τ_Q are the spontaneous emission lifetimes of the KrF*(B) state, that of the KrF*(C) state, the characteristic time for the stimulated emission of KrF*(B–X), and the total collisional quenching lifetime, respectively. The quantities τ_v and τ_{BC} are the characteristic time for the vibrational relaxation and mixing of the KrF*(B) state and KrF*(C) state, respectively. Also, R is the KrF*(B,C) formation rate. We assumed the formation rate for the KrF*(B) and KrF*(C) states is the same value.[41] When the mixing rate is very fast, the ratio of N_B/N_C is characterized mainly by θ_B and θ_C. Therefore, the branching ratio of KrF* formation does not influence the KrF* number density in the KrF*(B) state and the KrF*(C) state. However, it cannot be thought that Eqs. (4)–(7) describe the exact physical processes of the vibrational energy transfer and/or the mixing processes. In order to treat the vibrational level as a descrete level, real relaxation processes can be expressed by the harmonic oscillator approximation including V–V and V–T processes. However, in Eqs. (4)–(7), the quantities of τ_v and τ_{BC} are unknown parameters. Moreover, using the same type of two-state, upper-level model for the vibrational relaxation, the saturation behavior of sidelight fluorescence could be explained by Jacob *et al.*[21] Using the same type of two-state, upper-level model for the XeCl*(B) and (C) mixing, the pressure dependence of the XeCl*(B–X) and the (C–A) fluorescences for e-beam-excited XeCl laser could be explained by Kannari *et al.*[27] Therefore, we employ this four-state, upper-level model as an acceptable model of the upper laser level. The pair values of τ_v and τ_{BC} obtained in this paper is the quantities valid only for this four-state, upper-level model.

In our model, as the accessible vibrational levels of the KrF*(B) state, we took the levels $0 \leqslant v \leqslant 4$. This assumption is based on the calculation of Einstein's A coefficient for the KrF*(B–X) transition reported by Morgan and coworkers.[26] Their calculation showed that the vibrational populations in $0 \leqslant v \leqslant 4$ can all be extracted efficiently. Higher vibrational levels than $v = 4$ can contribute to the stimulated emission less efficiently. However, the vibrational quantum number dependence on its accessibility makes the model complex and also the consideration of their gain spectrum widths is necessary. Therefore, for simplicity, we ignored the contribution of higher vibrational levels than $v = 4$ to the stimulated emission. Because the quantity of characteristic time constant of τ_v depends on the vibrational quantum number considered, the τ_v obtained in this paper is only valid for the model with the accessible levels of $0 \leqslant v \leqslant 4$. The quantities of θ_B, θ_C, and θ_{v_i} are calculated by the energy gaps of $E_B - E_C = 80\ \text{cm}^{-1}$ and $\Delta E_v = 327\ \text{cm}^{-1}$ ($= E_{v=4}/4$) and the instantaneous gas temperature.

D. Adjustment of collisional quenching rate

The collisional quenching rate constants of KrF* can be obtained experimentally as a product of the spontaneous lifetime and the collisional quenching rate constant.[42–45] Therefore, determination of their rate constants depends on the spontaneous lifetime. Quenching rate constants for F_2, 2Kr, 2Ar, and Kr + Ar measured by Jacob *et al.*[42,43] were calculated using the spontaneous lifetime of the KrF*(B) state of 6.5 ns calculated by Dunning and Hay.[46] The electron quenching rate constant measured by Trainor and Jacob[45] was also obtained by the same calculation. However, as mentioned in Sec. C, as a result of mixing of the KrF*(B) state with the (C) state, the effective spontaneous lifetime is longer than that of the KrF*(B) state only. Strictly speaking, the ratio of the KrF* number density in the B state and the C state depends on the gas pressure due to the finite mixing rate. Therefore, the determination of the quenching rate of KrF* by the Stern–Volmer plots method is not valid because of the assumption of the constant spontaneous lifetime of KrF*. Fortunately, however, since the energy separation $E_B - E_C = 80\ \text{cm}^{-1}$ between the KrF*(B) state and the (C) is smaller than the kT ($209\ \text{cm}^{-1}$) at room temperature (300 K), one can treat their mixing processes at a rate of a gas kinetic collision frequency. Therefore, the ratio of the KrF* number density at the B state and the C state can be determined by the Boltzmann distribution of the gas temperature. The effective spontaneous lifetime of the KrF*(B,C) is calculated to be 15.18 ns with the respective spontaneous lifetime of the B state and the C state being 6.99 and 75.19 ns.[47,48] The modified quenching rates can be calculated from the Stern–Volmer plots using this effective spontaneous lifetime. For ArF*, the same modification was done. We reexamined the collisional quenching rate constants using the individual spontaneous lifetimes of $\tau_B = 4$ ns and $\tau_C = 44$ ns,[47,48] and the Boltzmann factor determined by the energy level gap of $E_B - E_C = 120\ \text{cm}^{-1}$.[40] For example, the rate for the substitutive reaction of ArF* + Kr→KrF* + Ar, which is the prime pumping reaction of KrF*, decreases the previous rate constant[42] by 41.8%.

The three-body collisional quenching rates are gas-temperature dependent. As a result of the reexamination of the

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quenching rate constants, the values of three-body quenching rates do not agree well with the theoretical quenching rates of KrF*(B) reported by Shui *et al.*[49-51] Therefore, we used the experimental value $k\tau$ [(quenching rate constant) $\times$(spontaneous lifetime)] obtained by Mangano *et al.*[43] and Jacob *et al.*,[42] and the rate constant was treated here as a temperature dependent rate, which was based on the theoretical calculation by Shui *et al.*[49-51]

When we introduce both the vibrational relaxation in the KrF*(B) and (C) states with the finite rates, the accessible KrF* number density to the stimulated emission at 248 nm wavelength becomes less than that calculated by the one-state, upper-level model. But modifying the collisional quenching rates can cause the KrF* molecules lying at the C state and the higher vibrational level of the B state to lase; otherwise they cannot contribute to the stimulated emission. Therefore, the small-signal gains do not result in a drastic decrease relative to those obtained by the one-state, upper-level model. At the early stage of KrF laser research, the fluorescence efficiencies and the fluorescence yield of KrF* have been measured by Nakano *et al.*[52] and Mangano *et al.*,[43] and good agreement was obtained with the prediction of the steady-state analysis using a reliable set of reaction rate constants. In their analyses, the quenching rates appeared only as a product of the quenching rate constant and the spontaneous lifetime of KrF*. Therefore, the adjustment of quenching rate constants using the effective spontaneous lifetime does not reduce good agreement between the experimental and theoretical results. In fact, our computer analysis using the adjusted rate constants showed good agreement with their fluorescence measurement.

III. SIDELIGHT FLUORESCENCE CHARACTERISTICS

We measured the sidelight fluorescences in the presence and absence of the cavity flux of 248-nm wavelength to determine the vibrational relaxation rate.

A schematic of the KrF laser excited transversely by e-beams is shown in Fig. 1. The e-beam diode operated at 660 kV with a peak current of 30 kA and pulse width of 50 ns. The excitation area is 40×4 cm^2. The excitation rate is 0.75 MW/cm^3 atm. The e-beam diode and the laser cell are immersed in the axial magnetic field produced by a pair of magnetic coils. For completely uniform excitation and precise determination of the excitation area, a strong magnetic field of 5 kG is applied. Under these conditions, the incident e-beam cross-sectional pattern detected by a rad-color film[53] was uniform and its area corresponded to the electron-emitting cathode area. The e-beam deposition energy was carefully checked by an e-beam calorimeter and a pressure-jump method in N_2 gas. The laser resonator in this experiment consisted of a 10-m-radius-of-curvature concave, 7.5-cm-diameter, dielectric-coated reflector and flat, 10-cm-diameter, dielectric-coated output couplers with various reflectivities (20%,40%,60%, and 80%). Both optical windows on the laser cell were 7.5-cm-diameter, CaF_2 flats without antireflection coating and were set perpendicularly to the optical axis. The percent transmission of each window was 60% at 248 nm. Sidelight fluorescences were measured from the sidelight viewing port located at the center of an optical axis. Full sidelight came from the intracavity region, because the strong axial magnetic field was applied. Sidelight was measured by a biplanar phototube ($S-5$) through a optical filter ($\lambda_0 = 250$ nm, $\Delta\lambda = 14$ nm FWHM). The degree of sidelight depletion was measured as a function of the cavity flux strength which was varied by the reflectivities of the output coupler and for parameters of laser gas pressure. The fluorescence in the absence of cavity flux was measured by masking the laser windows. The reflectivity of the windows could also dominate the lasing process. Therefore, in our model, we took the window reflectivity effect into account. The laser gas mixture was kept constant at Ar/Kr/F_2 of 89.8%/10%/0.2%, and was refilled by every shot. Laser output energy was measured by a Scientech 36-0401 calorimeter. The time histories of sidelight fluorescence at a total pressure of 2 atm for various reflectivities of the output coupler are shown in Fig. 2. Similar results were obtained at total pressures of 1, 1.5, and 3 atm. Because the magnetic field was not strong enough to suppress the scattered e-beam at 3 atm, we deleted those data from consideration. In the case of such short-pulse excitation as our 50-ns FWHM e-beam pulse, the steady-state analysis done by Jacob *et al.*[20,21]

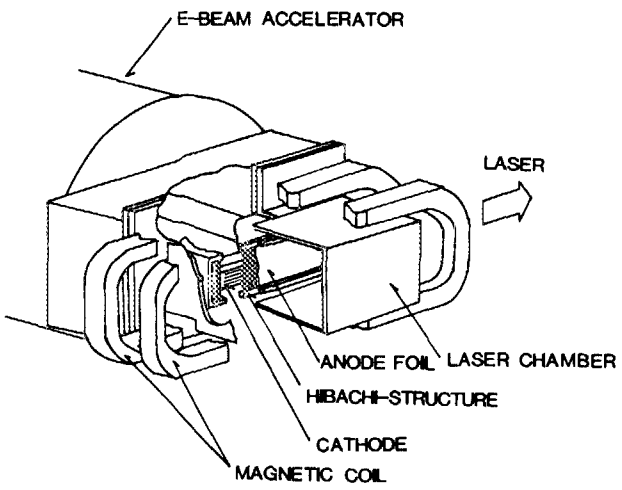


FIG. 1. Schematic of a transversely electron-beam-excited KrF laser.

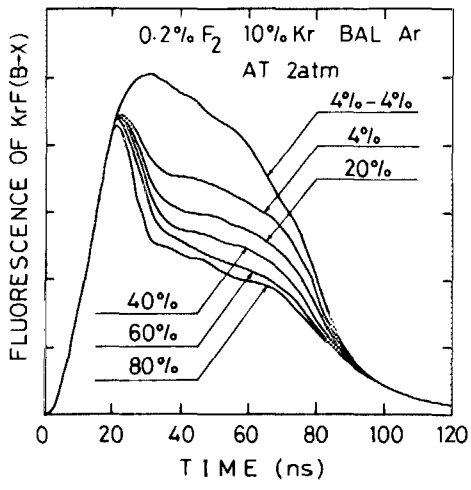


FIG. 2. Experimental time histories of KrF*(B–X) sidelight fluorescence at a total pressure of 2 atm for various reflectivities of the output coupler.

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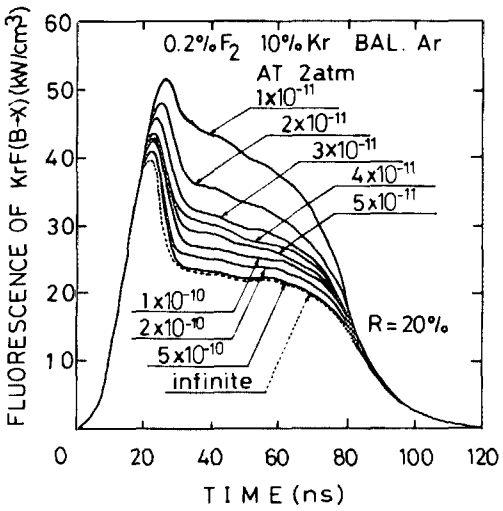


FIG. 3. Calculated time histories of KrF*(B–X) sidelight fluorescence for a reflectivity of the output coupler of 20% at a total pressure of 2 atm. The parameter is the vibrational relaxation rate constant. The mixing rate constant of 5×10^{-10} cm^3/s is used.

cannot be applied. Therefore, we compared the temporal behavior of the fluorescence pulse with the one calculated by the computer model. Because the used model was one dimensional, the time history of the sidelight detected from the center of the optical axis could be calculated. The calculation contains the transmission of laser windows and their reflectivities, the absorption by F_2 in the nonexcitation region, and the intracavity propagation time.

The calculated results for the reflectivity of the output coupler of 20% at a total pressure of 2 atm are shown in Fig. 3 for parameters of the vibrational relaxation rate constant. The mixing rate constant of 5×10^{-10} cm^3/s was used. The magnitude of the fluorescence at plateau was sensitive to the vibrational relaxation rate constant. The profile of the initial

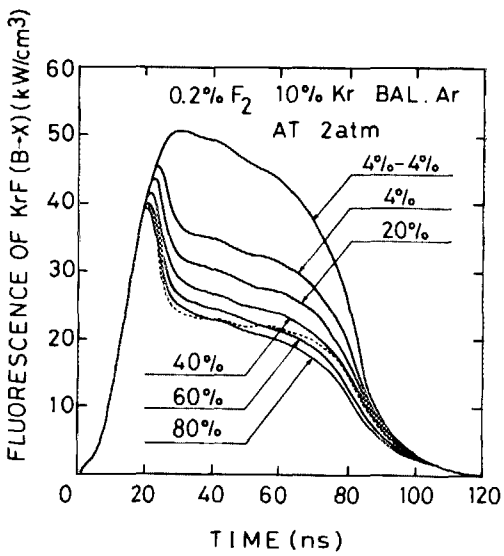


FIG. 4. Calculated time histories of KrF*(B–X) sidelight fluorescences at a total pressure of 2 atm for various reflectivities of the output coupler. The vibrational relaxation rate constant of 4×10^{-11} cm^3/s and the mixing rate constant of 5×10^{-10} cm^3/s are used. The time history shown by the dashed line is calculated at the infinite vibrational relaxation rate for the reflectivity of 20%.

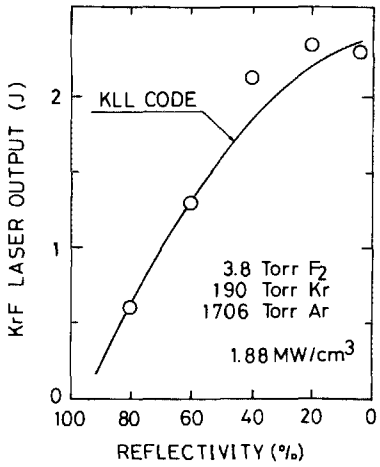


FIG. 5. Comparisons of the laser output energy between our code prediction and the experiment as a function of the reflectivity of the output coupler. The total laser gas pressure is 2.5 atm. The e-beam excitation rate is 1.88 MW/cm^3.

peak before the beginning of laser oscillation was varied by the KrF*(B)/(C) mixing rate constant. The best agreement between the calculated and measured time history was obtained at a vibrational relaxation rate constant of 4×10^{-11} cm^3/s. The calculated results at a total pressure of 2 atm are shown in Fig. 4 for various output couplers. The vibrational relaxation rate constant of 4×10^{-11} cm^3/s and the mixing rate constant of 5×10^{-10} cm^3/s were used. By comparing the calculated with measured time histories at total pressures of 1 and 1.5 atm, this pair of rate constants gave us the best agreement. In the calculation, the Boltzmann factor was varied during excitation by using the instantaneous laser gas temperature. This effect was, however, not appreciable in our excitation-rate regime.

Comparisons of the laser output energy between the model and the experiment as a function of reflectivity are shown in Fig. 5. The same rate constants as mentioned above for the vibrational relaxation and the KrF*(B)/(C) mixing are used here. This good agreement shown in Fig. 5 supports the sidelight depletion analysis, because the magnitude of the depletion is very sensitive to the cavity flux. Calculation was also performed for the parameters reported by Jacob *et al.*,[21] and the degree of sidelight depletion as a function of the cavity flux is shown in Fig. 6 along with Jacob's experimental results. Using the constant excitation rate against the

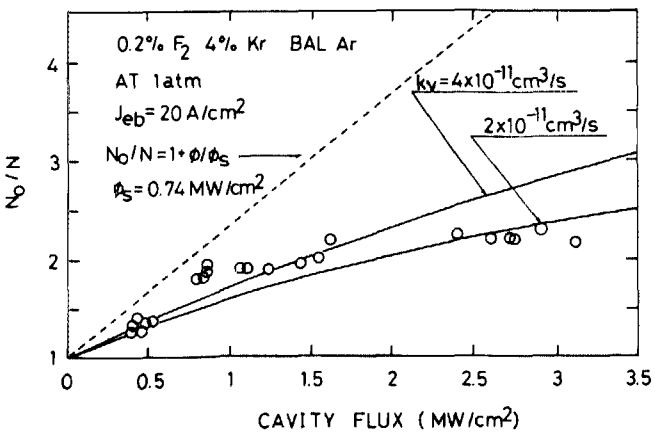


FIG. 6. Comparisons of the sidelight fluorescence depletion ratio between our code prediction and the experiment reported by Jacob *et al.* (see Ref. 21) as a function of cavity flux. The parameter is vibrational relaxation rate constant. The mixing rate constant of 5×10^{-10} cm^3/s is used. The dashed line shows the predicted behavior for infinite vibrational relaxation rate.

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TABLE II. Comparisons of small-signal absorption coefficient (%/cm) between our code prediction and experimental result reported by Klimek *et al.* (Ref. 17).

J_{eb} (A/cm^2)	1		2		3		Pressure (Amagats)
	0.2	0.6	0.2	0.6	0.2	0.6	F_2(%)
110	0.86	0.73	1.8	1.3	2.2	1.6	Exp.
	0.83	0.62	1.67	0.91	2.00	1.41	Code
55	0.41	0.32	0.91	0.60	1.1	0.89	Exp.
	0.48	0.37	0.97	0.53	1.11	0.82	Code
20	0.21	0.18	0.42	0.26	0.53	0.38	Exp.
	0.21	0.15	0.37	0.24	0.44	0.31	Code

time, fluorescence intensity was determined under the steady-state conditions. Calculated results are compared for the vibrational relaxation rate constants of 4×10^{-11} and 2×10^{-11} cm^3/s. The dashed line in Fig. 6 indicates the saturation profile of the sidelight calculated by using a infinite vibrational relaxation rate. The saturation intensity under this experimental condition is 0.74 MW/cm^2. By comparing the experimental result with the calculated result, fairly good agreement could be obtained for cavity fluxes less than 2 MW/cm^2 at the vibrational relaxation rate constant of 4×10^{-11} cm^3/s. Over prediction of the sidelight depletion at higher cavity fluxes in excess of 2 MW/cm^2 cannot be explained to date. In comparison of fluorescence time histories between our model predictions and our experimental results, and in comparison between the simulation results and experimental results reported by Jacob *et al.*, we finally determined the vibrational relaxation rate constant of 4×10^{-11} cm^3/s for argon. The mixing rate constant of KrF*(B) and (C) could not be determined accurately. However, as mentioned above, its rate should be fast because of their small energy level separation ($\Delta E < kT$). The sidelight fluorescence measurements with the cavity flux reported by Pummer *et al.*[24] in He buffer showed the fast mixing behavior ($\tau_{BC} \leqslant 1$ ns). In Ar buffer, the mixing rate will be faster than in He buffer. The difference between the laser output performance calculated with an infinite KrF*(B)/(C) mixing rate and modified collisional quenching rate constants, and that with a finite mixing rate and previous quenching rate constants ignoring the KrF(C) state is quite large. But variation of the KrF*(B)/(C) mixing rate constant in the rate range of 1×10^{-9}–1×10^{-10} cm^3/s does not have considerable influence on the intrinsic laser efficiency. Using the fast mixing rates in excess of 5×10^{-10} cm^3/s, the laser performance became nearly the same as that calculated under the statistic mixing conditions. This fact supports the validity of our adjustment for the collisional quenching rate under the statistic mixing conditions.

IV. SMALL-SIGNAL GAIN, ABSORPTION, AND INTRINSIC EFFICIENCY

In terms of the sidelight depletion with the cavity flux, our new model predicted well the experimental results. In this chapter, the validity of our new model is also examined by comparing the small-signal gain, the absorption coefficient, and the intrinisic laser efficiency with those experimental results reported by some authors. The small-signal gains measured by Klimek *et al.*,[17] for various excitation rates and laser gas parameters, are summarized along with the corresponding our code prediction in Table II. In these calculations, we used constant excitation rates against time. The corresponding *e*-beam current densities are shown. Hence, the small-signal gains are under steady-state conditions. As mentioned previously, the small-signal gains were determined from the total KrF* number density at the vibrational levels of $0 \leqslant v \leqslant 4$ in the KrF*(B) state. The small-signal absorption is listed in Table III. These values represent the total absorption in the laser mixture, except the contribution

TABLE III. Comparisons of small-signal gain (%/cm) between our code prediction and experimental result reported by Klimek *et al.* (Ref. 17).

J_{eb} (A/cm^2)	1		2		3		Pressure (Amagats)
	0.2	0.6	0.2	0.6	0.2	0.6	F_2(%)
110	9.1	8.3	14	11	15	11	Exp.
	9.5	10	14	13	14	16	Code
55	5.1	5.3	6.6	5.1	6.6	5.3	Exp.
	6.3	5.6	8.1	6.6	7.7	7.1	Code
20	1.9	1.4	2.3	1.6	2.2	1.4	Exp.
	2.1	1.6	2.3	1.7	2.0	1.5	Code

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by F_2. As can be seen from the results in Tables II and III, coincidence between the experiment and our new model prediction is very close for wide operational parameters. It should be noted that in spite of the considerable adjustment of the quenching rates for ArF* and KrF*, and the novel four-state, upper-level treatment, the model prediction was found quite precise. And our calculated results also agree well with the one-state, upper-level model reported by Klimek *et al.*[17] in terms of the small-signal gain and absorption. These facts mean that our novel model involving a four-state upper level changes only the energy extraction mechanism, and does not change the excitation mechanism. Similar comparisons of the small-signal gain and absorption were carried out with the experimental results reported by Edwards *et al.*,[54] Hawryluk *et al.*,[55] and Tisone *et al.*[5–7] The net-gain predictions agree well with the experimental results reported by Edwards *et al.* But the code predictions for the small-signal absorption were by 30%–40% lower than the experiment. There is no reasonable explanation for this discrepancy at present. Agreement between the experimental results reported by Hawryluk *et al.*[55] and our model is very fine. Our code predictions for the small-signal gain at the excitation conditions for experiments reported by Tisone *et al.*[5–7] were found to be higher by 20% for the 1000-Torr mixture and by 100% for the 2500-Torr mixture. Then, in comparison of our model with their experimental results, we calculated variation of the laser output power as a function of output coupling, which corresponds to the figure in Ref. 5 in which they determined the small-signal gains and small-signal absorptions. The prediction is shown in Fig. 7 along with the experimental results shown in Ref. 5. For the 1000-Torr mixture agreement is very good, while for the 2500-Torr mixture the code prediction for the laser power seems to be slightly lower than the experiment. It should be noted that in spite of the different set of small-signal gain, absorption, and saturation intensity, variation of the laser power with the output coupling is nearly the same. We have to know that it is difficult to get the unique solution from a Rigrod curve-fitting method. The sets of our model prediction for small-signal gain, small-signal absorption, and saturation intensity are also shown in Fig. 7. The excitation rate by Tisone *et al.*[5–7] is very high and the ion number density in the laser mixture during excitation is in excess of 10^{14} cm^{-3} at which the influence of the plasma shielding effect on the ion-ion recombination reaction is cannot be neglected. As can be seen in Fig. 7, the agreement of laser power between our predictions and experiments supports the validity of introduction of the plasma-shielding effect into our model. In addition to the small-signal gain and absorption, a number of experimental parameters shown by various authors is presented in Table IV. The excitation pulse shapes used for calculation are not real ones reported by various authors. We used the same pulse shape in all cases only by adjusting their FWHM. Considering this point, the agreement with experiments is fairly good. In the experiment by Edwards *et al.*, the intrinsic efficiency of 11% was achieved by injecting the external laser flux to help the buildup of the laser. Therefore, the reduction of the intrinsic efficiency in our model is mainly due to the slow buildup of the laser which reduces the extraction efficiency in the short excitation regime.

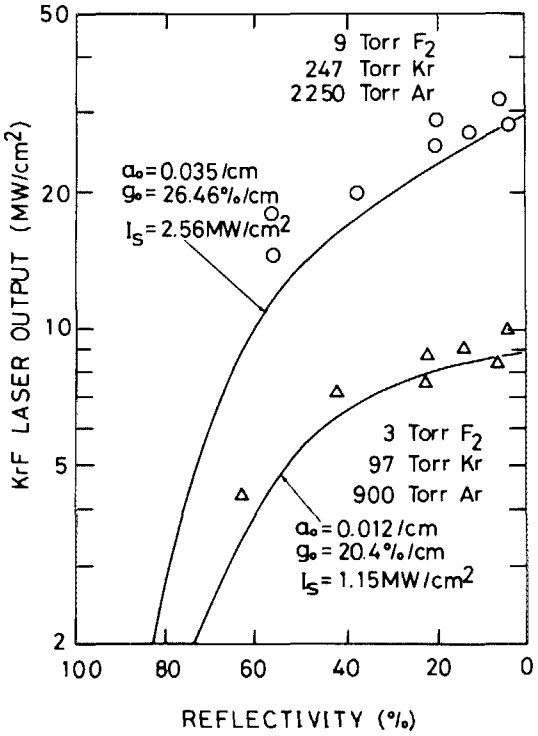


FIG. 7. Comparisons of the laser output power between our code prediction and the experiment reported by Tisone *et al.* (see Ref. 5) as a function of the reflectivity of the output coupler. The sets of code prediction for small-signal gain, small-signal absorption and saturation intensity are shown in the figure.

V. PICOSECOND DYNAMICS OF KrF FLUORESCENCE AND GAIN

Recently, Corkum and Taylor[56] have reported the picosecond gain dynamics of a discharge-excited XeCl amplifier measured by a double-pulse method using high-intensity, picosecond laser and probe pulses. They determined the double-exponential gain-recovery time constants of 40 ps and 2.5 ns. In the case of XeCl lasers of the bound-bound transition of B–X, the dissociation mechanism of the lower state (X) influences the gain recovery phenomena. They claimed that the short time constant of 40 ps is attributed to the dissociation rate constant of XeCl(X) and the rotational rate constant in the XeCl(B) state, and the long time constant of 2.5 ns comes from the balance condition with the effective lifetime of the XeCl(B) state. In the case of the KrF laser, the lower state is bound free. Therefore, it is important to measure the picosecond recovery-time constants for 248-nm gain and fluorescences (B–X and C–A) which reflect the relaxation mechanism and mixing mechanism. Recently, Szatmári and Schäfer[25] have measured the picosecond gain dynamics in a discharge-excited KrF amplifier using He buffer. From their results, only one recovery-time constant of 4 ns was obtained. They concluded that the time constant resulted from the contributions of the vibrational relaxation in the KrF*(B) state and the (C) state and of the repumping by the discharge electrons. The gain recovery dynamics in the Ar buffer gas would be different from the ones in the He buffer

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TABLE IV. Code predictions of the KrF laser oscillator.

	Experimental data					*KLL* code
	Pulse width (ns)	Gain length (cm)	Total pressure (atm)	Excitation rate (MW/cm^3)	η_{int} (energy)	η_{int} (energy)
Ewing *et al.*[58]	50	50	1.5	0.7	9	10.8
Ewing *et al.*[58]	50	50	2	1.2	9	10.8
Tisone *et al.*[5–7]	50	40	1.3	1.8	13	10.1
Tisone *et al.*[5–7]	50	40	3.3	7	10	8.1
Edwards *et al.*[8]	60	35	2	1.5	11	8.6
Jacob *et al.*[20]	600	100	1.7	0.18	9(power)	10.4

gas because the relaxation rate and the mixing rate by the Ar gas collision are higher than those by the He gas.

In this chapter, we calculated the picosecond gain dynamics using our four-state, upper-level model for parameters of the vibrational relaxation rate and the KrF*(B)/(C) mixing rate. Then, we predicted the possible information which will be obtained through a series of recovery dynamics measurements for the gain and the fluorescence. The excitation is kept constant through entire pulse at an excitation rate of 2 MW/cm^3 for the 2-atm mixture of Ar/Kr/F_2 = 89.8%/10%/0.2%. After the 50-ns constant excitation, the laser pulse with a temporal profile of $I_0\exp(-t/\tau)$, where I_0 denotes the peak laser intensity and τ is the characteristic pulse width, was injected. We calculated the recovery dynamics of 248-nm gain, KrF*(B–X) fluorescence, KrF*(C–A) fluorescence, and Kr_2F^* fluorescence. Those time histories calculated using a vibrational relaxation rate constant of 4×10^{-11} cm^3/s and a mixing rate constant of 5×10^{-10} cm^3/s for the incident peak intensity of $I_0 = 500$ MW/cm^2 and characteristic time constant of $\tau = 5$ ps, are shown in Fig. 8. Under these excitation conditions used in this calculation, the value of 500 MW/cm^2 corresponds to its saturation intensity. Therefore, the saturation energy density is 2.5 mJ/cm^2. As can be seen in Fig. 8, the fast-recovery time constant of about 0.1 ns for the KrF*(B)/(C) mixing and the slow-recovery time constant of about 7 ns appeared in both the 248-nm gain and the KrF*(B–X) fluorescence. In the KrF*(C–A) fluorescence, only a slow-recovery time constant could be found.

Variation of the recovery dynamics with the vibrational relaxation rate is shown in Fig. 9. Figure 9(a) is for vibrational relaxation rate constant of 4×10^{-10} cm^3/s and Fig. 9(b) is for 4×10^{-12} cm^3/s. Because the KrF*(B)/(C) mixing rate is kept constant, we can see the same fast recovery characteristics as shown Fig. 8. A decrease in the gain with decreasing the vibrational relaxation rate is attributed to the fact that during lasing the number density of the KrF* lying at higher vibrational levels above $v = 4$ increases at a slow vibrational relaxation rate. From Figs. 8, 9(a), and 9(b), even when changing the vibrational relaxation rate constants, there is no considerable change either in the fast-recovery time constant or in the slow-recovery constant. Therefore, we cannot determine the vibrational relaxation rate solely from the gain recovery observations. However, the modulation depth of the gain at the laser pulse injection increases with increasing the vibrational relaxation rate. Therefore, if one initially knows the exact peak power of the injected laser pulse, we can estimate the vibrational relaxation rate from the measurement of the modulation depth of the 248-nm gain dip.

Variation of the recovery dynamics with the KrF*(B)/(C) mixing rate is shown in Fig. 10. Figure 10(a) is for the mixing rate constant of 5×10^{-11} cm^3/s and Fig. 10(b) is for 5×10^{-12} cm^3/s. Here, the vibrational relaxation rate constant is kept constant at 4×10^{-11} cm^3/s. The fast-recovery time constant increases proportionally to the inverse of the

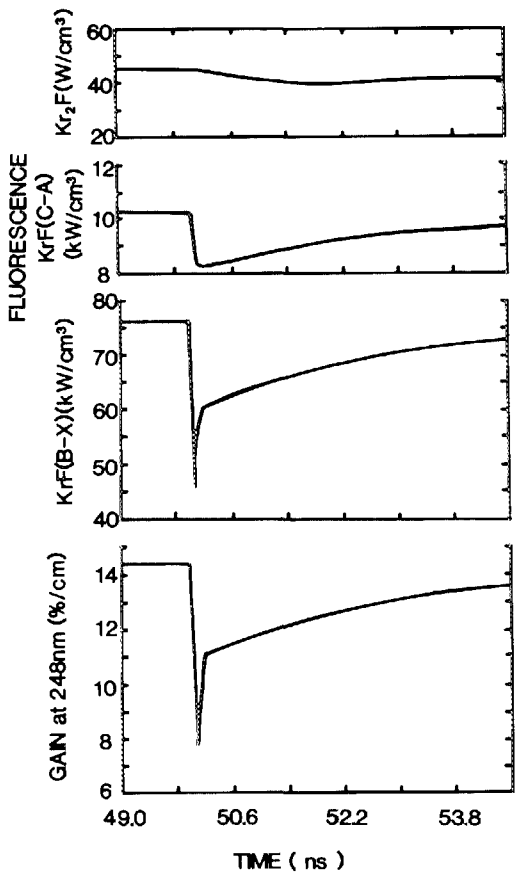


FIG. 8. Code predictions for picosecond recovery dynamics of small-signal gain (B–X) and fluorescences of KrF*(B–X), (C–A), and Kr_2F^* with a vibrational relaxation rate constant of 4×10^{-11} cm^3/s and a mixing rate constant of 5×10^{-10} cm^3/s. The laser gas mixture is Ar/Kr/F_2 = 89.8%/10%/0.2% at a total pressure of 2 atm. The excitation rate is 2 MW/cm^3. The temporal profile of the injected KrF laser is $I_0\exp(-t/\tau)$, where $I_0 = 500$ MW/cm^2 and $\tau = 5$ ps.

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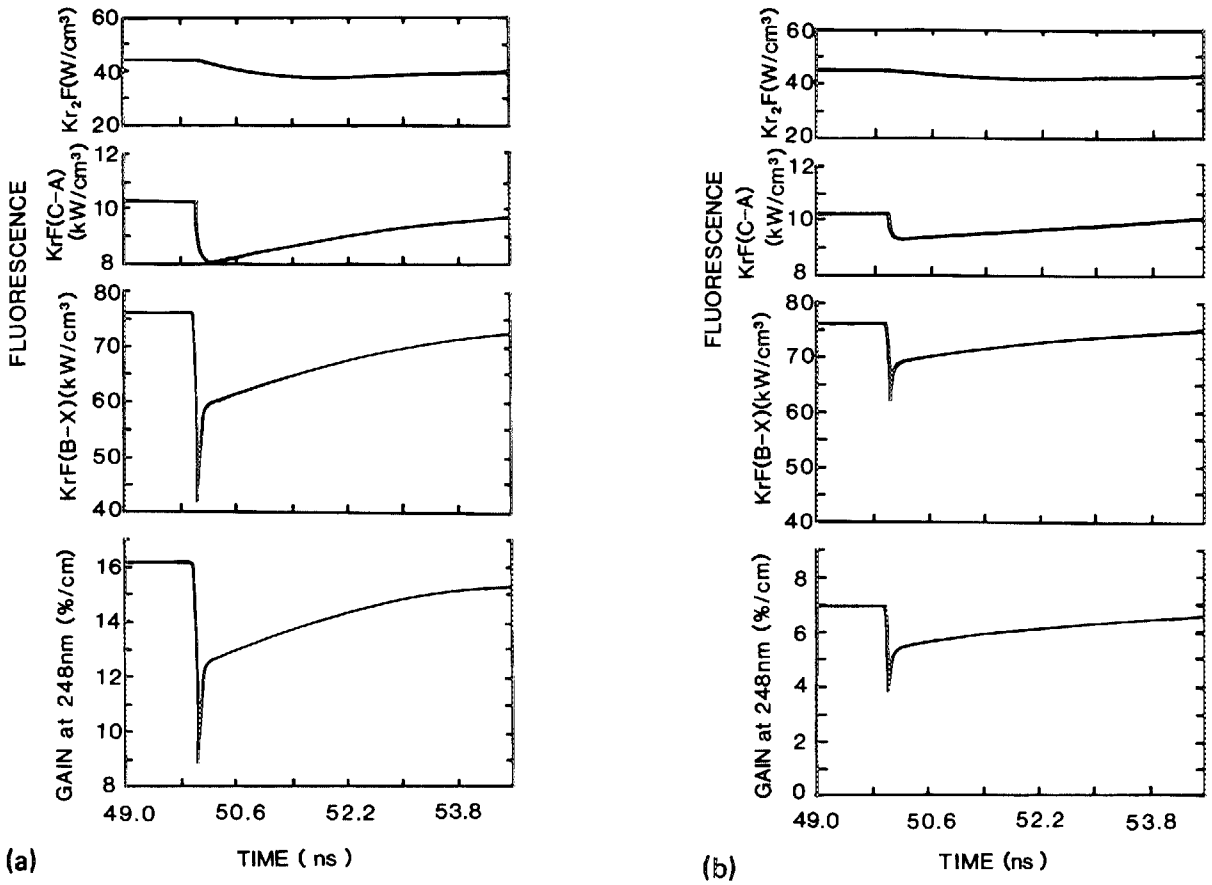


FIG. 9. (a) Code prediction for picosecond recovery dynamics of small-signal gain and fluorescences of KrF*(B–X), (C–A), and Kr_2F^*. Calculated conditions are the same as those of Fig. 8 except the vibrational relaxation rate constant of 4×10^{-10} cm³/s. (b) Code predictions for picosecond recovery dynamics of small-signal gain and fluorescences of KrF*(B–X), (C–A), and Kr_2F^*. Calculated conditions are the same as those of Fig. 8 except the vibrational relaxation rate constant of 4×10^{-12} cm³/s.

mixing rate. At the mixing rate constant of 5×10^{-12} cm³/s, the gain dynamics can be characterized solely by the slow-recovery time constant. In the case of the KrF*(C–A) fluorescence, the modulation depth increases with decreasing the mixing rate. Therefore, it is relatively easy to estimate the KrF*(B)/(C) mixing rate from the recovery dynamics of the gain and the fluorescence. The experimental results reported by Szatmári and Schäfer[25] in the He buffer look like the recovery dynamics shown in Fig. 10(b). However, our calculated results do not support their estimation of the time constant of KrF*(B)/(C) mixing, because the influence of the mixing rate on the modulation depth of the 248-nm gain dip is negligible. The slow-recovery time constant reflects the effective storage time of KrF*(B,C), being nearly constant. Therefore, the estimation of the vibrational relaxation time constant from the slow-recovery time constant is not valid. In the experiment by Szatmári and Schäfer,[25] the modulation depth of 248-nm gain is nearly 100%. However, due to the lack of the information of peak laser intensity, we cannot estimate the vibrational relaxation rate for their excitation parameters.

Variation of the recovery dynamics with the injected laser intensity is also calculated, those results of which are not presented. As a result, for the recovery time measurement of the 248-nm gain and the fluorescences, the laser pulses of $\tau\leqslant50$ ps and $I_0\geqslant50$ MW/cm² were found to be necessary.

VI. CONCLUSIONS

The kinetic computer models developed so far on e-beam-excited KrF lasers that include the vibrational relaxation in the upper laser level could not predict the higher intrinsic efficiency due to the reduction of the laser extraction efficiency. We have developed a four-state, upper-level model. On the assumption of fast mixing of the KrF*(B) state with the (C) state due to the small energy differences of $\Delta E=E_B-E_C<kT$, the effective spontaneous lifetime becomes over twice as long as the lifetime for the KrF*(B) only. We have reexamined the collisional quenching rates using this effective spontaneous lifetime of the KrF*(B,C) state. By comparing time histories of sidelight fluorescence when loading the cavity flux in our experiments using a 50-ns FWHM e-beam with our new model predictions, we estimated the vibrational relaxation rate constant and the KrF*(B)/(C) mixing rate constants as 4×10^{-11} and 5×10^{-10} cm³/s, respectively, for a two-body collision rate constant with Ar gas. Our new model using these rate constants can predict the saturation behavior of KrF*(B–X) fluorescences, small-signal gains, small-signal absorptions, and intrinsic efficiencies reported by various authors without any adjustment factors. In terms of the small-signal gain, the absorption and the intrinsic efficiency, the values predicted by our four-state, upper-level model are the same as those calculated by the one-state, upper-level model that ne-

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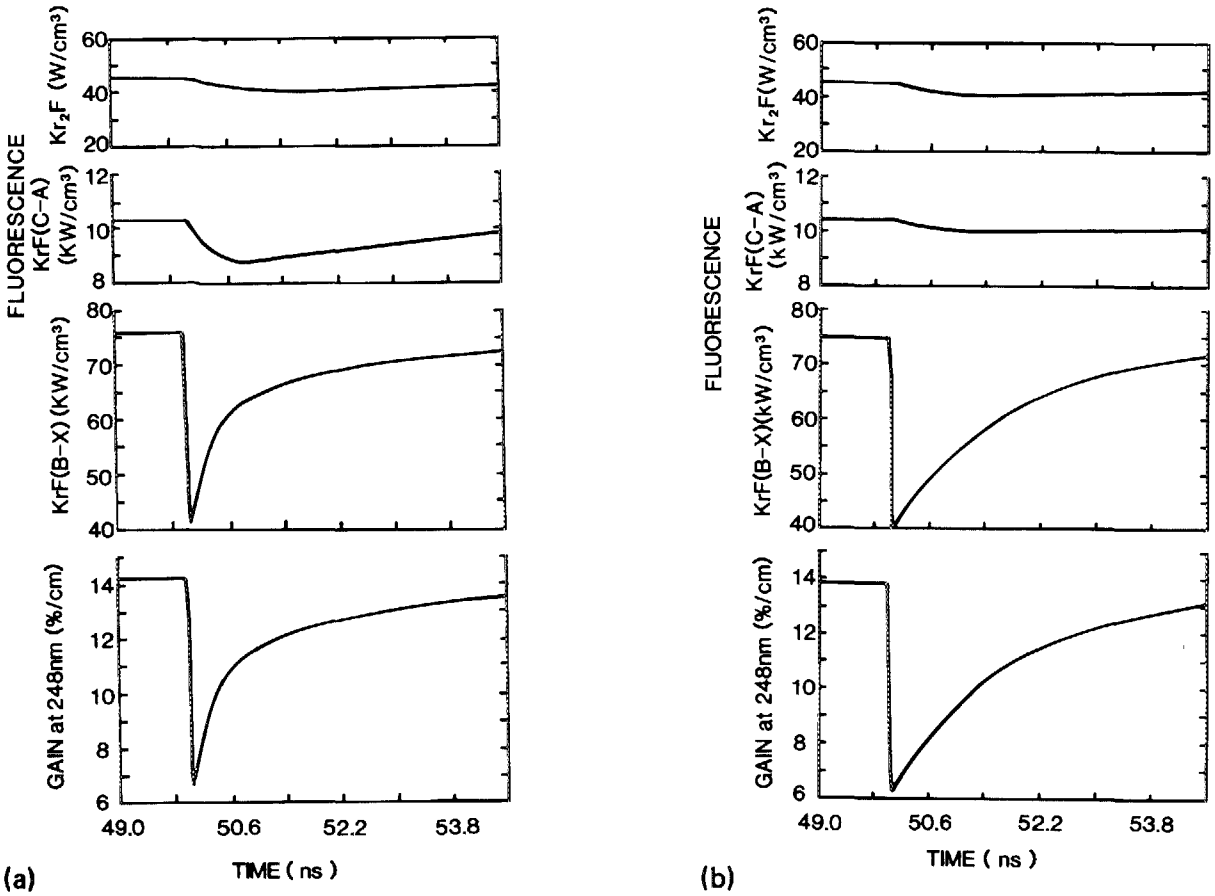


FIG. 10. (a) Code predictions for picosecond recovery dynamics of small-signal gain and fluorescences of KrF*(B–X), (C–A), and Kr_2F*. Calculated conditions are the same as those of Fig. 8 except the mixing rate constant of 5×10^{-11} cm³/s. (b) Code predictions for picosecond recovery dynamics of small-signal gain and fluorescences of KrF*(B–X), (C–A), and Kr_2F*. Calculated conditions are the same as those of Fig. 8 except the mixing rate constant of 5×10^{-12} cm³/s.

glects the vibrational relaxation and the KrF*(B)/(C) mixing processes.

Further information on the vibrational relaxation and the KrF*(B)/(C) mixing is obtainable by measuring picosecond recovery dynamics of 248-nm gain and KrF*(B–X) and (C–A) fluorescences using a picosecond, high-power laser pulse. Because the fluorescence of KrF*(C–A) overlaps in spectrum the Ar_2F* fluorescence in the Ar buffer, the modulation depth measurement and the fast-recovery time-constant measurement of the 248-nm gain will be helpful for estimating the vibrational relaxation rate and the KrF*(B)/(C) mixing rate. The measurement of the slow-recovery time constant reflects only the storage time of the upper level of KrF*.

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